universal joints (A & B) are there in between differential unit and wheel, movable coupling (C) allows to move up and down motion of the wheel. A shock absorber is incorporated to the system. With the help of these components, the system absorb the road disturbance when the vehicle passes over the bumps and pits.

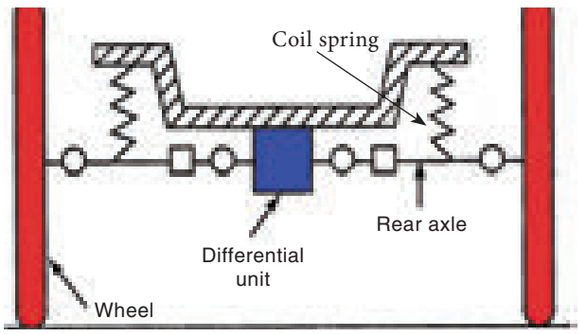


Figure 7.3.2.6 Rear axle independent suspension system

7.3.3 Air suspension system

These types of suspension systems are commonly used in a long-distance vehicle like buses. Due to the benefits of the air suspension system, nowadays this system is widely used in many vehicles.

Schematic representation of this type of air suspension system is shown in the figure 7.3.3(1&2). It consists of four air springs which may be a bellow type or piston type. Air compressor is used to pressurize the purified air (through filter) from the atmosphere and stored in the accumulator at a pressure maintained between 5.6 to 7 kg/cm^2. To release the excess air and to maintain the pressure inside the accumulator, a safety relief valve is used. The high pressure compressed air enters into the air spring through leveling valve and lift control valve. The lift control valve was operated by a manual lever which is located in the panel board.

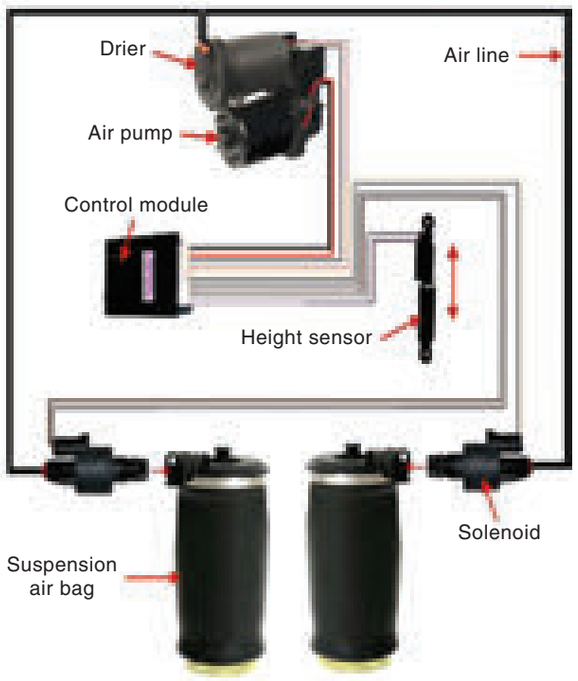


Figure 7.3.3 (1) Air suspension system

Advantages

1. It reduces noise, vibration, and harshness due to which passenger and driver attain better ride comfort. It reduces the journey tiredness of the driver and the passenger.
2. Change of spring deflection is quite low when compared with a conventional suspension system for both loading and unloading condition. This helps to reduce the load transfer on the vehicle.
3. Headlamp alignment remains constant though the vehicle is travelling over irregular roads.
4. With the help of automatic controlling tools, this system will provide some space for required wheel movement.

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